

L10



L10: The number of positive divisors ①
function

Definition: Let $n \in \mathbb{Z}$ with $n > 0$. The number of positive divisors function, denoted $\nu(n)$, is the function defined by

$$\nu(n) := |\{d \in \mathbb{Z} : d > 0 \text{ and } d|n\}|.$$

Theorem 3.6: The number of positive divisors function is multiplicative.

Proof: Observe that

$$\nu(n) = \sum_{d|n, d>0} 1.$$

Since the constant arithmetic function $f(d) = 1$ is multiplicative (check this yourself), we have that $\nu(n)$ is multiplicative by Theorem 3.1. $\square$

• Since $\nu(n)$ is multiplicative, it is determined by its values at powers of prime numbers.

Theorem 3.7: Let p be a prime number and let $a \in \mathbb{Z}$ with $a \geq 0$. Then $\nu(p^a) = a + 1$.

Proof: The positive divisors of p^a are precisely

$1, p, p^2, \dots, p^{a-1}, p^a$. There are exactly $a+1$ such divisors. $\square$

Theorem 3.8: Let $n = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$ with $p_1, p_2, \dots, p_r$ distinct prime numbers and $a_1, a_2, \dots, a_r$ nonnegative integers. Then

$$V(n) = \prod_{i=1}^r (a_i + 1).$$

Proof: We have

$$\begin{aligned} V(n) &= V(p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}) \\ &= V(p_1^{a_1}) V(p_2^{a_2}) \dots V(p_r^{a_r}) \text{ by Theorem 3.6} \\ &= (a_1 + 1)(a_2 + 1) \dots (a_r + 1) \text{ by Theorem 3.7} \\ &= \prod_{i=1}^r (a_i + 1). \end{aligned} \quad \square$$

Example: Consider $504 = 2^3 3^2 7$. Then

$$V(504) = (3+1)(2+1)(1+1) = 24.$$

The sum of positive divisors function

Definition: Let $n \in \mathbb{Z}$ with $n > 0$. The sum of positive divisors function, denoted $\sigma(n)$, is the function defined by

$$\sigma(n) := \sum_{d|n, d>0} d.$$

Theorem 3.9: The sum of positive divisors function is multiplicative. (3)

Proof: Since the arithmetic function $f(d) = d$ is multiplicative (check this yourself), we have that $\sigma(n)$ is multiplicative by Theorem 3.1. $\square$

• Since $\sigma(n)$ is multiplicative, it is determined by its values at powers of prime numbers.

Theorem 3.10: Let p be a prime number and let $a \in \mathbb{Z}$ with $a \geq 0$. Then

$$\sigma(p^a) = \frac{p^{a+1} - 1}{p - 1}.$$

Proof: The positive divisors of p^a are precisely $1, p, p^2, \dots, p^{a-1}, p^a$. Using the geometric series formula we obtain,

$$1 + p + p^2 + \dots + p^a = \frac{p^{a+1} - 1}{p - 1}.$$

$\square$

Theorem 3.11: Let $n = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$ with $p_1, p_2, \dots, p_r$ distinct prime numbers and $a_1, a_2, \dots, a_r$ non-negative integers. Then

$$\sigma(n) = \prod_{i=1}^r \frac{p_i^{a_i+1} - 1}{p_i - 1} \quad (4)$$

Proof: We have

$$\begin{aligned} \sigma(n) &= \sigma(p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}) \\ &= \sigma(p_1^{a_1}) \sigma(p_2^{a_2}) \dots \sigma(p_r^{a_r}) \text{ by Theorem 3.9} \\ &= \left(\frac{p_1^{a_1+1} - 1}{p_1 - 1} \right) \left(\frac{p_2^{a_2+1} - 1}{p_2 - 1} \right) \dots \left(\frac{p_r^{a_r+1} - 1}{p_r - 1} \right) \\ &\quad \text{by Theorem 3.10} \\ &= \prod_{i=1}^r \frac{p_i^{a_i+1} - 1}{p_i - 1}, \end{aligned}$$

as required. 

Example: Consider $504 = 2^3 3^2 7$.

$$\begin{aligned} \text{Thus } \sigma(504) &= \left(\frac{2^4 - 1}{2 - 1} \right) \left(\frac{3^3 - 1}{3 - 1} \right) \left(\frac{7^2 - 1}{7 - 1} \right) \\ &= 1560. \end{aligned}$$

Perfect numbers

Definition: Let $n \in \mathbb{Z}$ with $n > 0$. Then n is said to be a perfect number if $\sigma(n) = 2n$.

• If $n \in \mathbb{Z}$ with $n > 0$, then $\sigma(n) - n$ is the (5)
sum of all the positive divisors of n other than n itself. Thus we have the following equivalent definition of a perfect number.

Definition: Let $n \in \mathbb{Z}$ with $n > 0$. Then n is said to be a perfect number if $\sigma(n) - n = n$, or equivalently, if the sum of all positive divisors of n other than n itself is equal to n .

Example: • 6 is a perfect number since

$$\sigma(6) = 1 + 2 + 3 + 6 = 12 = 2 \cdot 6$$

• 28 is a perfect number since

$$\sigma(28) = 1 + 2 + 4 + 7 + 14 + 28 = 56 = 2 \cdot 28.$$

Theorem 3.12: Let $n \in \mathbb{Z}$ with $n > 0$. Then n is an even perfect number if and only if

$$n = 2^{p-1}(2^p - 1)$$

where p is a prime, and $2^p - 1$ is also a prime (i.e. $2^p - 1$ is a Mersenne prime).

• Theorem 3.12 gives a characterisation of

even perfect numbers and a bijection between (6) Mersenne Primes and even perfect numbers?

We don't know if there are infinitely many Mersenne primes or not. There are 38 known even perfect numbers, each corresponding to a Mersenne prime. The first five even perfect numbers correspond can be found using $p = 2, 3, 5, 7$ and 13 in Theorem 3.12.

Conjecture: There are infinitely many even perfect numbers.

Conjecture: Every perfect number is even.

Proof (of Theorem 3.12): ($\Rightarrow$ due to Euler). Assume that n is an even perfect number. Write $n = 2^a b$ with $a, b \in \mathbb{Z}$ with $a \geq 1$ and b odd. Then,

$$\begin{aligned}\sigma(n) &= \sigma(2^a b) = \sigma(2^a) \sigma(b) = \left(\frac{2^{a+1} - 1}{2 - 1} \right) \sigma(b) \\ &\quad \text{by Theorem 3.9} \\ &= (2^{a+1} - 1) \sigma(b).\end{aligned}$$

Since n is perfect, we have

$$\sigma(n) = 2n = 2(2^a b) = 2^{a+1} b.$$

(7)

Thus,

$$(2^{a+1} - 1)\sigma(b) = 2^{a+1}b. \quad (1)$$

Now $2^{a+1} \mid (2^{a+1} - 1)\sigma(b)$, and thus $2^{a+1} \mid \sigma(b)$

Since $(2^{a+1}, 2^{a+1} - 1) = 1$. Thus

$$\sigma(b) = 2^{a+1}c \quad (2)$$

For some $c \in \mathbb{Z}$. Substituting (2) into (1), we obtain

$$(2^{a+1} - 1)2^{a+1}c = 2^{a+1}b,$$

or equivalently,

$$(2^{a+1} - 1)c = b. \quad (3)$$

We now prove that $c = 1$. Assume, by way of Contradiction, that $c > 1$. Then by (3), b has at least three distinct positive divisors: $1, b, c$ (we have $2^{a+1} - 1 > 1$ since $a \geq 1$). Thus

$$\sigma(b) \geq 1 + b + c.$$

But from (2) we have,

$$\sigma(b) = 2^{a+1}c = (2^{a+1} - 1)c + c = b + c.$$

This is a contradiction! Thus $c = 1$, from which (3) gives $b = 2^{a+1} - 1$, and (2) gives

$\sigma(b) = 2^{a+1} = b+1$. The latter equation (8) implies that b is a prime number, from which

$$n = 2^a b = 2^a (2^{a+1} - 1),$$

where $2^{a+1} - 1$ is a prime number, and so $a+1$ is also a prime (by the Homework).

($\Leftarrow$ direction, due to Euclid): Assume that $n = 2^{p-1}(2^p - 1)$ where p and $2^p - 1$ are prime numbers. Then

$$\sigma(n) = \sigma(2^{p-1}(2^p - 1))$$

$$= \sigma(2^{p-1}) \sigma(2^p - 1) \text{ by Theorem 3.9}$$

$$= \left(\frac{2^p - 1}{2 - 1}\right) 2^p \quad (\text{since } 2^p - 1 \text{ is prime by hypothesis}).$$

$$= (2^p - 1) 2^p$$

$$= 2 \cdot 2^{p-1} (2^p - 1)$$

$$= 2n,$$

as required. □

Möbius function

Definition: Let $n \in \mathbb{Z}$ with $n > 0$. The Möbius

(9)

μ -function, denoted $\mu(n)$, is given by

$$\mu(n) = \begin{cases} 1, & \text{if } n=1 \\ 0, & \text{if } p^2 \mid n \text{ with } p \text{ prime} \\ (-1)^r, & \text{if } n = p_1 p_2 \cdots p_r \text{ with} \\ & p_1, p_2, \dots, p_r \text{ distinct prime} \\ & \text{numbers.} \end{cases}$$

Example: Since $504 = 2^3 3^2 7$, we have that $\mu(504) = 0$.

• Since $30 = 2 \cdot 3 \cdot 5$ we have $\mu(30) = -1$.

Theorem 3.13: The Möbius function $\mu(n)$ is multiplicative.

Proof: Let m and n be relatively prime positive integers. We must show that $\mu(mn) = \mu(m)\mu(n)$. If $m=1$ or $n=1$, the desired result is clear. Now, m or n is divisible by the square of a prime if and only if mn is divisible by the square of a prime number (since $(m,n)=1$). In this case, both sides of the desired result reduce to 0. Assume that $m = p_1 p_2 \cdots p_r$ and $n = q_1 q_2 \cdots q_s$

with $p_1, p_2, \dots, p_r, q_1, q_2, \dots, q_s$ being distinct primes. Then (10)

$$\begin{aligned}\varphi(mn) &= \varphi(p_1 p_2 \dots p_r q_1 q_2 \dots q_s) \\ &= (-1)^{r+s} \\ &= (-1)^r (-1)^s \\ &= \varphi(m) \varphi(n),\end{aligned}$$



As required.

Proposition 3.14: Let $n \in \mathbb{Z}$ with $n > 0$. Then

$$\sum_{d|n, d>0} \varphi(d) = \begin{cases} 1 & \text{if } n=1 \\ 0 & \text{otherwise.} \end{cases}$$

Proof: The arithmetic function

$$F(n) = \sum_{d|n, d>0} \varphi(d)$$

is multiplicative by Theorem 3.13 and Theorem 3.1. Thus F is determined by its values at prime powers. Clearly, $F(1) = 1$. If p is a prime and $a \in \mathbb{Z}$ with $a > 0$, then

$$F(p^a) = \sum_{d|p^a, d>0} \varphi(d)$$

$$= \varphi(1) + \varphi(p) + \varphi(p^2) + \dots + \varphi(p^{a-1}) + \varphi(p^a) \quad (11)$$

$$= 1 + (-1) + 0 + \dots + 0$$

$$= 0,$$

as desired. □

Example: Let $n = 12$. Then

$$\sum_{d|12, d>0} \varphi(d) = \varphi(1) + \varphi(2) + \varphi(3) + \varphi(4) + \varphi(6) + \varphi(12)$$

$$= 1 + (-1) + (-1) + 0 + 1 + 0$$

$$= 0.$$